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TradePilot – Bear-Day Audit, Regime Engine & Learnings

Session audit log + comprehensive learnings – 2026-06-08 to 2026-06-09

Project	TradePilot (paper-trading engines)
Scope	Trade-audit tooling · v7_regime engine · long/short research · data hygiene
Branch	<code>dev</code> (10 commits)
Trigger	Bear day (2026-06-08): v4 long-only bled, v5 short-tilted profited
Status	Complete – v7 live as paper A/B from 2026-06-09 08:45 IST
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Executive summary

A bear session (Nifty -0.94%) exposed a structural flaw: TradePilot's scorer emits **zero SELL signals**, so the long-only engine (v4) piled 47 longs into a falling market and bled, while the short-tilted engine (v5) made money. We built a per-trade **counterfactual audit** that quantified the leak (Rs 52,221 left on the table; `LONG_IN_BEAR` alone $-\text{Rs } 18,826$), ran **adversarially-verified research** into long/short/flip systems, and shipped a **two-layer regime engine (v7_regime)** – a daily ADX/DMI gate that sets the *allowed side*, plus an intraday Supertrend flip within it.

The honest twist: **walk-forward optimization showed the daily gate has no historical edge** (Deflated Sharpe 0.12). Thresholds aren't the lever; the engine's real bet is the *intraday* flip, which has no history to backtest. So v7 went live as a **paper experiment** – the only valid way to test it – graded automatically each day by the audit tool. We also fixed 22 rotted NSE tickers in the universe and added a rot-guard.

METRIC	VALUE	NOTE
Realized P&L (2026-06-08)	–Rs 11,128	194 trades, 121 long / 73 short
Profit left on the table	Rs 52,221	recoverable with right side + timing
Largest leak (<code>LONG_IN_BEAR</code>)	–Rs 18,826	bigger than the whole net loss
Daily-gate WFO edge	none (DSR 0.12)	thresholds are not the fix
Universe tickers repaired	22 of 29	verified vs live Yahoo before applying
Commits shipped	10	all on <code>dev</code> , tests green

Session audit log (chronological)

#	PHASE	WHAT HAPPENED	OUTCOME
1	Launch	Started the market stack for 2026-06-08 open	3 engines + dashboard up
2	Triage	“v4 not scanning” → found it’s the intentional 09:30 yfinance warm-up	not a bug
3	Audit build	Built <code>trade-audit.py</code> – 3-layer per-trade counterfactual	root cause found: 0 SELL signals
4	Observe	Started missed-opportunities + profit watchdogs (were never launched)	session recorded live
5	Diagnose	EOD audit: –Rs 11,128, Rs 52k on table, <code>LONG_IN_BEAR</code> dominant	quantified the leak
6	Research	Deep-research harness: 23 sources, 16 claims confirmed / 9 killed	two-layer spec
7	Build	v7_regime engine via TDD + subagent-driven dev	Layer 1 gate + Layer 2 flip, 10 tests
8	Intraday	Wired Layer 2 to live 5-min candles (daily fallback)	true intraday flip, 13 tests
9	Data	Refreshed daily CSVs (452 symbols)	NIFTY-500 current to 2026-06-08
10	Validate	Walk-forward tuning + Deflated Sharpe	honest: no historical edge
11	Go live	v7 = 4th engine, paper A/B, armed for 08:45 auto-launch	first live evidence today
12	Hygiene	Resolved 22 renamed NSE tickers + rot-guard	universe 390/397 fresh

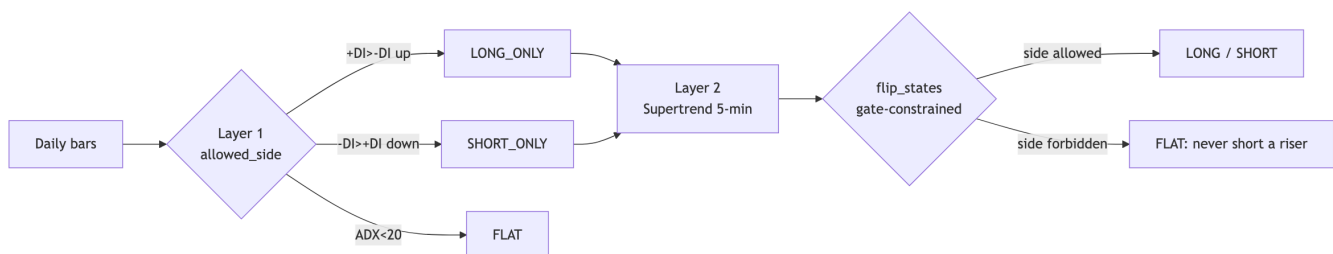
The diagnosis – why the bear day bled

The audit classified every trade. One mistake class explains the entire loss:

MISTAKE CLASS	COUNT	REALIZED	ON TABLE
LONG_IN_BEAR	78	−Rs 18,826	Rs 37,650
GOOD_TRADE	57	+Rs 8,245	Rs 6,258
EXIT_TOO_EARLY	35	+Rs 994	Rs 5,205
SHORTED_RISER	22	−Rs 1,535	Rs 3,068
IGNORED_SIGNAL	2	−Rs 7	Rs 40

The natural experiment proved the fix before we built it: **v4 (long-only) −Rs 8,325; v5 (short-tilted, 40 shorts) +Rs 465** on the same falling day. The problem was never stock-picking – it was the absence of a regime gate and a SELL signal.

What we built – the two-layer regime engine



2026-06-09_session-audit-and-learnings-diagram-0

MODULE	LINES	PURPOSE
<code>prototype/v7/regime_gate.py</code>	~70	Layer 1: ADX/+DI/-DI + SMA50 slope → allowed_side
<code>prototype/v7/supertrend_flip.py</code>	~70	Layer 2: Supertrend SAR + gate-constrained flip machine
<code>scripts/v7_regime-paper-trade.py</code>	(v5 clone)	the engine – direction swapped for gate+flip on 5-min bars
<code>scripts/v7-regime-backtest.py</code>	~60	Layer-1 gated-vs-buy&hold backtest
<code>scripts/v7-wfo-tune.py</code>	~150	walk-forward ADX tuning + Deflated Sharpe
<code>scripts/trade-audit.py</code>	~360	per-trade counterfactual + bear-day solution
<code>scripts/fix-stale-tickers.py</code>	~140	universe rename resolver + rot-guard

Design intent: Layer 1 (slow, daily) decides *which side is even allowed*; Layer 2 (fast, intraday 5-min) decides *when to enter and flip* within it. A side the regime forbids collapses to FLAT – making “short a riser / long a faller” structurally impossible. 13 unit tests pass; the smoke gate stays green.

The honest validation result



Walk-forward optimization across the 49-stock NIFTY-50 basket found **no ADX-threshold combination with a positive out-of-sample Sharpe** (best ≈ -0.10), and a Deflated Sharpe Ratio of **0.12** – essentially no evidence of skill. We did **not** fabricate a tuning win. v7 ships with 25/20 (least-bad) explicitly labelled *experiment, not validated edge*. Because the intraday Layer-2 flip cannot be backtested (no intraday history), the **forward paper A/B is the only honest validation path** – and the audit grades it daily.

Learnings

#	LEARNING	WHY IT MATTERS
1	The leak was structural, not stock-picking	No SELL signal + no regime gate → bear days are a long-only bloodbath regardless of which stocks are chosen
2	Counterfactual auditing beats P&L watching	"Rs on the table" + mistake-class breakdown points at the fix; raw P&L only says you lost
3	Single-symbol backtests lie	TCS "+6.6% vs -60%" was regime-cherry-picking; basket WFO + DSR exposed zero edge
4	DSR is the honesty check	A good-looking Sharpe across many tried variants is usually selection luck; DSR 0.12 said "no"
5	Some theses can't be backtested	Layer-2 needs intraday history that doesn't exist → forward paper is the only valid test; ship it labelled as an experiment
6	Hardcoded universes rot	29 NSE tickers had renamed/demerged (ZOMATO→ETERNAL, TATAMOTORS demerger); added a <code>--audit</code> rot-guard
7	Verify data mappings before applying	A wrong ticker loads the wrong company – worse than a gap; tested all 22 renames vs live Yahoo first
8	TDD pure modules first	Layers 1 & 2 are pure functions with tests; the engine just wires them – fast, safe, reviewable
9	Right side beats right timing	Flipping every wrong-direction trade would have added +Rs 40,718 vs +Rs 26k from better timing alone
10	Watchdogs must be in the launch path	The observers existed but were never started; capture only began once launched mid-session

Artifacts & commits

COMMIT	SUMMARY
<code>f4d000b</code>	trade-audit engine – per-trade counterfactual + bear-day solution
<code>295c4bb</code>	two-layer long/short/flip research spec
<code>b64979d</code>	directional indicators (ADX/+DI/-DI)
<code>0e11e45</code>	Supertrend SAR + gate-constrained flip machine
<code>1e92dc7</code>	Layer-1 walk-forward-lite backtest
<code>9f9d7cf</code>	regime-gated paper engine (opt-in)
<code>319472a</code>	Layer 2 flips on intraday 5-min candles
<code>76d3bdd</code>	walk-forward ADX tuner + Deflated Sharpe
<code>18aee44</code>	go live as paper A/B (4th engine)
<code>05e0554</code>	resolve 22 renamed tickers + rot-guard

Reports rendered: [docs/audit/2026-06-08_audit-report.pdf](#) , [docs/research/2026-06-08_long-short-flip-spec.pdf](#) , [docs/superpowers/plans/2026-06-08-v7-regime-engine.pdf](#) , and this document.

Open items / next steps

ID	ITEM	PRIORITY
NEXT-1	Review v7's first live A/B via tonight's auto-audit (15:35)	high
NEXT-2	If v7 whipsaws, add Supertrend flat-line stand-aside (spec chop guard)	medium
NEXT-3	Add a real SELL tier to the scorer (the actual root-cause fix)	high
NEXT-4	Re-fetch the 7 unresolved tickers periodically (Yahoo coverage gaps)	low
NEXT-5	Wire <code>fix-stale-tickers.py --audit</code> into the weekly cron	low

Bottom line: We turned a losing bear day into a quantified diagnosis, a researched fix, a shipped experiment, and a cleaner data foundation – while staying honest that the edge is unproven and today is the first real test.